# SPOORTHI BASU

**Software Engineer | Real-Time Data Infrastructure & Distributed Systems**
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## SUMMARY

Software engineer with 5 years building real-time data infrastructure and fault-tolerant distributed systems on Kafka, Flink, and Iceberg. Apache Flink CDC contributor, InfoQ author, Data Streaming Summit 2026 speaker, with work under review at PVLDB and CACM Practice.

## PROFESSIONAL EXPERIENCE

**Software Engineer, Genesys Cloud Services, March 2021 – Present**

- Designed and built **Flink** pipelines behind the company's first **data lakehouse**, flattening company-wide **Kafka** event streams into schema-driven **Iceberg/S3** datasets at **hundreds of millions of events per day**; recognized with the company's **Go Big Award** for the delivery.
- Architected the company's **first end-to-end data validation framework**, asserting correctness, completeness, compliance, and query performance across **Kafka**, **Flink**, and **Iceberg**, from source events through to what customers query; **adopted by 3 teams**, run continuously.
- Scaled fault-tolerant **Java/Kafka microservices** to **10M+ events per day** via partitioning and consumer rebalancing, cutting latency by **25%**.
- Engineered customer-facing **REST APIs** sustaining **5K+ RPS** with **Redis** cache sharding, halving database load during peak traffic hours.
- Built org-wide **integration testing 0→1** (**LocalStack**), cutting test creation to **under a minute** for 5+ teams, catching defects pre-release.
- Delivered **99.99% uptime** through **multi-region deployment** and circuit breakers (**Resilience4j**) for fault isolation across every service.
- Owned **on-call triage** for production data services with a **sub-5-minute P0/P1 response SLA**, diagnosing and fixing incidents **same day**.
- Mentored a junior engineer on distributed systems and ramped teammates onto the export testing project, halving onboarding time.

**Software Engineer, Coding Minds, July 2020 – February 2021**

- Shipped a full-stack academic platform (**React**, **Java**, **Node.js**) on Heroku, serving **500+ daily active users** across many courses and cohorts.
- Developed **RESTful APIs** (**Java/Spring**) over **MySQL**, hardened to **95% test coverage**, and cut page load time by 30% with memoization.

## RESEARCH, PUBLICATIONS, TALKS & OPEN SOURCE

**Speaker, Data Streaming Summit 2026, San Francisco (Oct 7–8)**

- *"Your Agent's Memory Is Only as Good as the Pipeline Behind It."* Selected through open CFP. Argues that exactly-once is a guarantee about processing, not what ends up in the table. An agent doesn't crash on wrong memory; it answers confidently and the error goes untraced.
- Grounds the argument in defects diagnosed and fixed upstream, including silent duplication under schema evolution and coordinator state retention on a 2.5B-row table, then turns to detection and the assertions a validation framework can make across Kafka, Flink, and Iceberg.

**Published, InfoQ: "The Schema Proliferation Problem in Kafka and Flink Pipelines"**

- Deep-dive that ranked **#1 Top Article** in InfoQ's weekly Software Development Round-Up, reviewed by a Java Champion before publication and read by a community of **550K+ senior developers** monthly, with an open-source reference implementation on GitHub.
- Diagnosed schema proliferation in **Kafka/Flink/Avro** pipelines as the cost of one-to-one event-to-schema mapping, then introduced a discriminator-based consolidation pattern using enum routing fields to retire the cross-table UNIONs it had forced.

**Apache Flink CDC, Open Source Contributions | FLINK-38450 (PR #4360) | FLINK-39775 (PR #4418)**

- PR #4360 (**merged, shipping in Flink CDC 3.7**): fixed duplicate records in the Iceberg sink when a schema change splits writes mid-checkpoint, committing each batch as its own snapshot so equality-deletes supersede stale rows. Committer-approved, with schema-change tests.
- PR #4418 (open): fixes a JobManager OOM on large MySQL CDC tables, ~300K finished splits for a 2.5B-row table, where the source coordinator retained snapshot-split metadata after entering the binlog phase and re-serialized it on every checkpoint thereafter.

**Under review, PVLDB | "Audit-Preserving Compaction for Merge-on-Read Tables" | Lean 4 proof + Iceberg patch | Artifact**

- Ships an audit-preserving compaction mechanism for **Apache Iceberg** that recovered **5,440 of 5,440** masked violations with zero false positives and zero misses, plus a machine-checked **Lean 4** development, a read-only checker, and a FLINK-38450 reproduction.
- Proves a merge-on-read table is faithful at every prefix exactly when physical ordering linearly extends logical version order, that the physical table alone cannot in general reveal this, and that no purely local ordering scheme can guarantee it.

**Under review, CACM Practice | "The History Was Already There" | Append-only audit pattern, deployed in production | Artifact**

- Reconstructs historical state at query time with SQL window functions, eliminating previous-value columns carried on every row: removing **67 such fields** cut schema definition size **48%** (1,415 to 734 lines) and per-event Avro binary size **38%** (539 to 334 bytes) in production.
- Solves the event-sequence problem so the reconstruction stays correct under clock skew and retries, and lifts historical depth from a hard cap of one hop to as far back as the data goes, with a runnable Apache 2.0 reference implementation on **Kafka**, **Flink**, and **Iceberg**.

## TECHNICAL SKILLS

- **Languages & Frameworks:** Java, Python, JavaScript, SQL, C++, Lean 4, Spring Boot (MVC, JPA)
- **Data & Streaming:** Apache Kafka, Apache Flink, Apache Iceberg, Apache Avro, CDC, Exactly-Once Semantics, Real-Time Data Pipelines
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, Athena, DynamoDB, SQS), Redis, Docker, Terraform, Jenkins, CI/CD, Multi-Region Deployment
- **Testing & Monitoring:** JUnit, Mockito, Testcontainers, LocalStack, Postman, New Relic, Sumo Logic, Integration Testing
- **Core Concepts:** Distributed Systems, System Design, Concurrency, Fault Tolerance, Formal Verification, Test-Driven Development
- **AI Development Tools:** Cursor, Claude Code, LLM-assisted code review and test generation

## EDUCATION

Master's in Computer Science, California State Polytechnic University, Pomona (May 2020) - GPA: 3.66
Bachelor's in Computer Science, Dr. Ambedkar Institute of Technology, Bengaluru, India (June 2018) - GPA: 4.0